

Colour evidence for a same-redshift quasar companion: the unified 41-band model (October 2026)

Internal method note, `qso_pcolor`

4 October 2026

Abstract

A quasar has a spectroscopic redshift z_0 , and a point source lies a few arcseconds away. This note describes how `qso_pcolor` measures, from broad-band photometry alone, how strongly the companion’s colours support the hypothesis that it is itself a quasar at z_0 . The method compares four populations: quasars at z_0 , quasars at other redshifts, everything else in the imaging catalogue (the background, mostly stars), and a broad catch-all. It accepts any subset of 41 bands from seven surveys that includes Legacy Surveys r at signal-to-noise ≥ 10 . One latent model, shared by the northern and southern Legacy Surveys, describes all bands, and photometry is corrected for Galactic extinction. Two complete models are provided on an equal footing. They share everything except the quasar colour model: in one, quasar colours do not depend on apparent magnitude (the XDQSO design); in the other, they may. The background is fitted, as in XDQSO, separately in eight bins of Legacy r , on a training sample from which likely quasars are removed, with component weights that vary smoothly with Galactic position. On held-out data the two models separate quasars from background objects with AUC 0.976 and 0.980.

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1 Summary of what is delivered

- **Two models on an equal footing** (bundles of 3 October 2026): **independent** (quasar colours independent of apparent magnitude) and **dependent** (quasar colours free to depend on it). Neither is the default; Section 6.2 explains the difference and when to prefer each, and we recommend scoring candidates with both. In `models/multisurvey_psf/work/stellar_binned/20261003/bundles/`. Load with `UnifiedPSFModel.load(...)`.
- **Inputs.** Any subset of 41 bands (SDSS *ugriz*; Legacy Surveys DR9 South and North *g, r, z, W1, W2*; AllWISE *W1–W4*; Pan-STARRS1 *grizy*; NSC *ugrizY+VR*; SkyMapper *uvgriz*; VHS *YJHK_s*) that contains Legacy *r* at $S/N \geq 10$, as catalogued fluxes or magnitudes with errors; candidate RA and Dec; the primary’s redshift z_0 ; morphology and blend information. Fainter sources, and sources without Legacy *r*, are flagged and not scored (Section 4.1).
- **Output.** A ranking statistic $\log \mathcal{R}$ (Section 3), evidence and posterior quantities, a quasar-support percentile, and explicit eligibility and quality flags.
- **Main results** (Section 10; first number for the magnitude-independent model, second for the magnitude-dependent one). AUC 0.976 and 0.980 on held-out quasars and background objects; 86% and 89% of quasars receive $p_{\text{quasar}} > 0.5$; 0.3% of background objects that are not flagged as likely quasars do. No artificial colour-grid point outside the models’ dense core receives $p_{\text{quasar}} > 0.5$. Among ranked objects p_{quasar} agrees with the observed quasar fraction in the mid range and is mildly overconfident above 0.97; about 3% of quasars are not ranked.

2 Notation and glossary

2.1 Symbols

Symbol	Meaning (units)
z_0	spectroscopic redshift of the primary quasar
z	redshift of a hypothetical companion quasar
$\mathbf{y}$	observed native luptitudes of a source in the bands it has (mag; Section 5)
$\mathbf{C}$	measurement covariance of $\mathbf{y}$ (mag^2 ; diagonal in practice)
$\mathbf{x}$	latent, noise-free luptitudes in the 36 shared coordinates (mag)
$H, \mathbf{b}, T$	observation operator: $\mathbf{y} = H\mathbf{x} + \mathbf{b} + \boldsymbol{\epsilon}$, with extra instrumental scatter $\boldsymbol{\epsilon} \sim N(0, T)$
m_{ref}	luptitude of the reference band, Legacy <i>r</i> , on which colours are conditioned (mag)
a	index of the reference band
E	SFD98 reddening $E(B - V)$ at the source position (mag)
R_j	extinction coefficient of band j , A_j/E
ℓ, b	Galactic longitude and latitude (deg)
$\Sigma_Q(z, m)$	quasar surface density per unit redshift and magnitude ($\text{deg}^{-2} \text{mag}^{-1}$)
$\Sigma_B(m)$	background surface density per unit magnitude ($\text{deg}^{-2} \text{mag}^{-1}$)
η	catch-all share of the background intensity (dimensionless)
λ_X	intensity of hypothesis X : surface density times colour likelihood
$\mathcal{R}$	ranking statistic, Eq. (3) (per unit redshift)
Δz_{eff}	effective width of the declared redshift window, <code>dz_match_eff</code>
K	number of Gaussian components in a mixture
w, ν	covariance-prior scale (10^{-3}mag^2) and strength (1)

2.2 Terms

Luptitude. The inverse hyperbolic sine magnitude of Lupton et al. (1999),

$$m = 22.5 - \frac{2.5}{\ln 10} \left[\sinh^{-1} \left(\frac{f}{2s} \right) + \ln s \right],$$

for flux f in the band's native unit and softening s . It equals the ordinary magnitude for $f \gg s$ and stays finite for zero or negative flux.

Native system. A band's own catalogue calibration. No band is converted into another survey's system.

Slice. The quasar model for redshifts within ± 0.1 of a centre z_k , for $z_k = 0.15, 0.25, \dots, 4.35$ (43 slices; neighbouring slices overlap).

Reference band. Legacy DR9 r : the southern band where observed, otherwise the northern one. Colour likelihoods are conditioned on it, and the background bins are defined in it. Every scored source has it at $S/N \geq 10$ (Section 4.1).

Role. Every training object belongs to one sky-based role: *fit* and *select* (training), *calibration* (catch-all shares and support threshold), or *test* (validation). Roles are fixed in advance and never reassigned.

Background. Every point source in the imaging catalogue that is not a quasar. It is dominated by stars and is fitted empirically. Its training sample excludes known quasars and likely quasars (Quaia members and WISE AGN colours; Section 7).

Catch-all. A broad Student- t density with a small share of the background intensity. It explains objects that neither the quasar nor the background mixture fits, so that Gaussian tails do not decide their classification.

Support percentile. For a candidate and z_0 , the fraction of simulated quasars at z_0 , with the candidate's own errors, whose conditional density is lower than the candidate's. A small value means the colours are unusual for a quasar at z_0 .

Held-out density. The mean, over test objects, of the conditional log density $\ln p(\mathbf{y}_{\text{other}} | m_{\text{ref}})$ (nat per object, density per mag^{N-1}). Higher is better.

AUC. The probability that a random test quasar ranks above a random test background object on $\log \mathcal{R}$. Objects the scorer declines to rank are placed at the bottom.

3 The question and the ranking statistic

Four hypotheses compete for a companion with photometry $\mathbf{y}$ near a primary at z_0 (Fig. 1):

1. $Q(z_0)$, a quasar at the primary's redshift, within a declared window;
2. $Q(z \neq z_0)$, a quasar at any other redshift;
3. B , a member of the background population;
4. U , the catch-all.

Each hypothesis has an intensity, a surface density times a likelihood. With reference magnitude m and conditional colour likelihoods,

$$\lambda_Q(z) = \Sigma_Q(z, m) p(\mathbf{y} | Q, z, m), \quad (1)$$

$$\lambda_B = (1 - \eta) \Sigma_B(m) p(\mathbf{y} | B, m, \ell, b), \quad \lambda_U = \eta \Sigma_B(m) p(\mathbf{y} | U, m). \quad (2)$$

We rank candidates on

$$\mathcal{R} = \frac{\lambda_Q(z_0)}{\int \lambda_Q(z) dz + \lambda_B + \lambda_U} \quad [\text{per unit redshift}]. \quad (3)$$

The denominator does not depend on the redshift window, because same-redshift and other-redshift quasars together form the whole quasar intensity. A posterior for a window of effective width Δz_{eff} follows by one multiplication, $p_{\text{same}} = \mathcal{R} \Delta z_{\text{eff}}$ (exact for the window-averaged $\mathcal{R}$ that the scorer stores). Photometric redshifts have widths of about $\sigma_z \approx 0.6$, while a $\pm 2000 \text{ km s}^{-1}$ window at $z = 1.4$ is $\Delta z = 0.016$ wide, so p_{same} is small even for an ideal candidate. $\log \mathcal{R}$ is the quantity to compare between candidates. Appendix A gives the derivation.

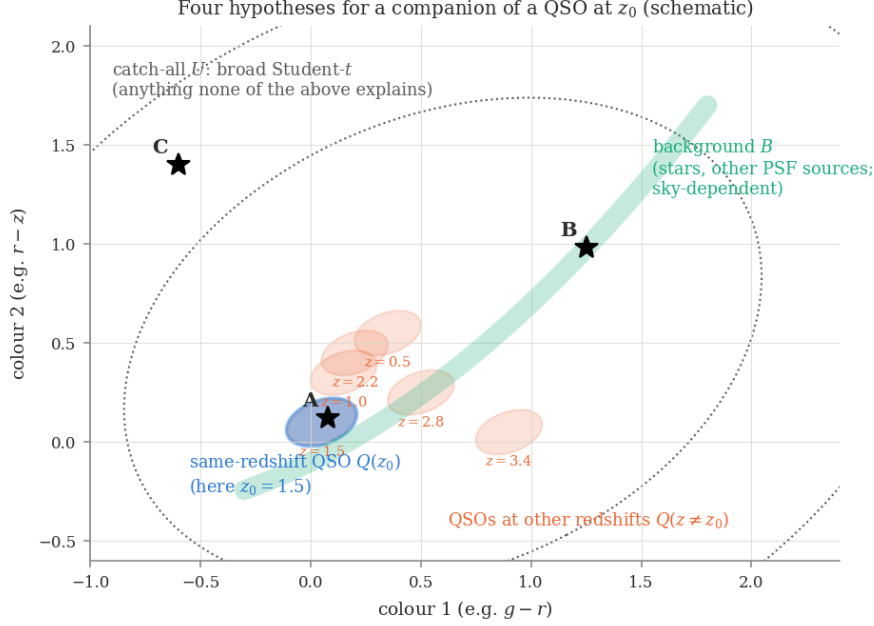


Figure 1: Schematic (synthetic, not fitted) of the four hypotheses in one colour plane. Candidate A lies on the same-redshift quasar locus, B on the background locus, and C far from both; the catch-all explains C, so the Gaussian tail of whichever population is wider does not decide it.

4 Data and roles

Quasars. The training quasars are the union of DESI DR1 quasars (`spectype=QSO`, `zwarn=0`, primary) and SDSS DR16Q (Lyke et al., 2020), merged by sky position. Of 1,657,444 eligible quasars, 1,049,260 are fit or select rows. Of these, 610,503 are DESI-only, 241,808 SDSS-only, and 196,949 in both (Fig. 4). The remaining quasars are 239,371 calibration and 368,813 test rows. Their photometry was acquired from all seven surveys and stored as native fluxes and variances.

Background. The background sample is 2,978,857 cleaned point sources in 274 sky regions (Legacy `type=PSF`, mask-clean, known quasars removed), with seven-survey photometry. Of these, 1,975,894 are fit or select rows and 502,251 calibration rows.

Selection. All samples satisfy $|b| \geq 25^\circ$. A band is used only when its survey marks it usable (finite flux, positive variance, a positive exposure count where reported, and the survey’s cleanliness flag); weak and negative fluxes are kept. The quasar models were trained without a signal-to-noise cut (0.7% of their fit/select rows lie below the faint limit). The background is trained, and all objects are scored, only above the faint limit of Section 4.1.

Roles. Roles are assigned by sky region before any fitting (Fig. 3). The background regions are 274 cones of radius 0.3° : 146 fit, 34 select, 45 calibration and 49 test cones, so test objects lie in separate sky regions. Rows used during development to choose stopping points or thresholds (34,259 quasars and 2,045 background objects for the quasar fits; the stopping and evaluation rows of the background design tests) are recorded and excluded from the tests in Section 10.

4.1 The faint limit

We train the background on, and score, only sources with Legacy r at $S/N \geq 10$ ($\sigma_r \lesssim 0.11$ mag; $\sigma(g-r) \approx 0.2$ mag). Below this the colour errors are as large as the colour difference between quasars and stars, and the model cannot vouch for a quasar. Measured before the limit was imposed (same quasar models, joint background), the fraction of held-out quasars given $p_{\text{quasar}} > 0.5$ is 19% at

$5 \leq S/N < 7$ and 30% at $7 \leq S/N < 10$, against 51% at $10 \leq S/N < 15$ and 93% above $S/N = 50$ (Fig. 2). The limit is a signal-to-noise cut rather than a magnitude cut because the two hemispheres differ in depth: stellar counts peak at $r \approx 24.6$ in the South and $r \approx 23.6$ in the North, and $S/N = 10$ falls at $r \approx 24.1$ and 22.9 respectively. It removes 2.8% of test quasars and 64% of test background objects, almost all faint stars. Legacy r is then the reference band of every scored source, so the quasar and background densities are conditioned on the same magnitude that defines the background bins. A source without Legacy r (for example SDSS-only photometry) is not scored; on the training samples that affects 0.01% of quasars.

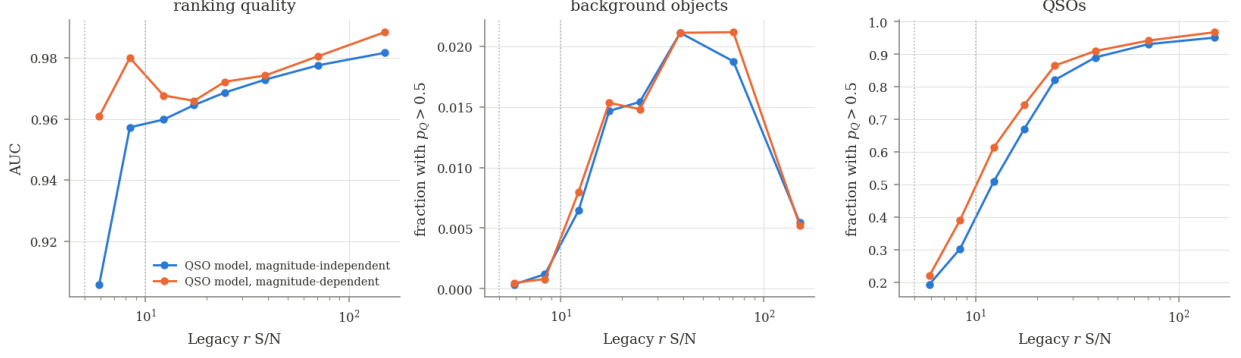


Figure 2: Held-out performance against Legacy r signal-to-noise, measured before the faint limit was imposed. Left: AUC within each S/N bin. Middle and right: fraction of background objects and of quasars given $p_{\text{quasar}} > 0.5$. Dotted lines: $S/N = 5$ and 10.

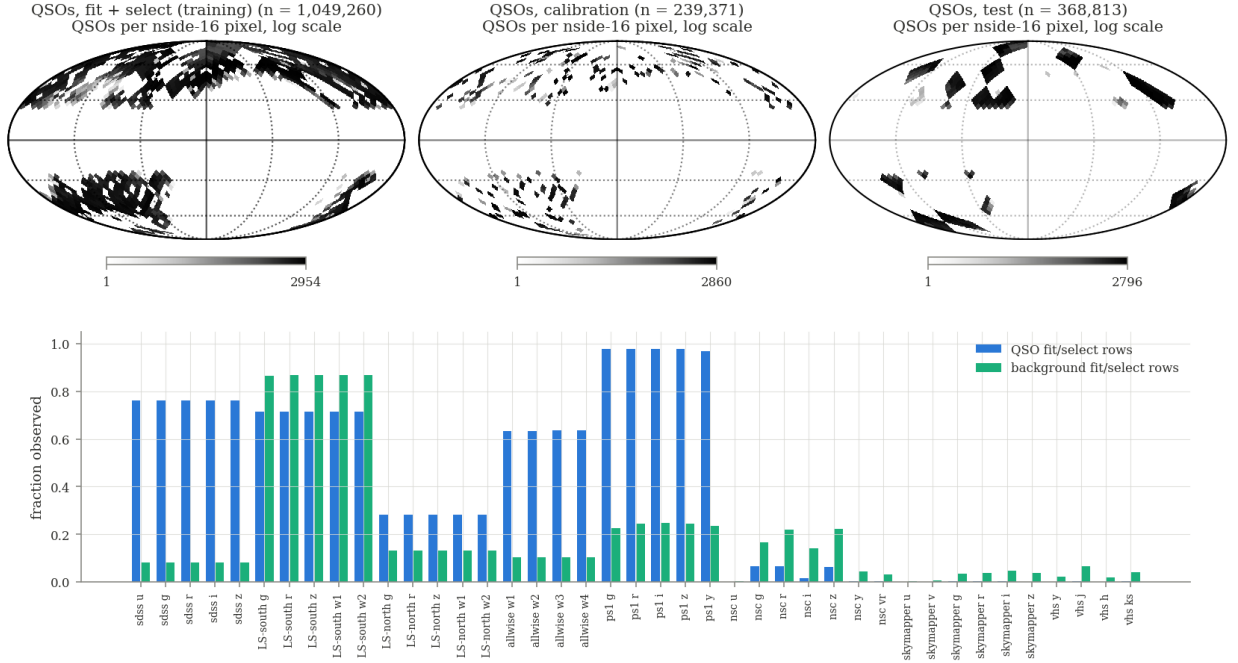


Figure 3: Top: sky distribution of quasars by role, Galactic Mollweide projection, counts per HEALPix nside-16 pixel (log scale). Bottom: fraction of fit/select rows with each band observed, for quasars (blue) and the background (aqua).

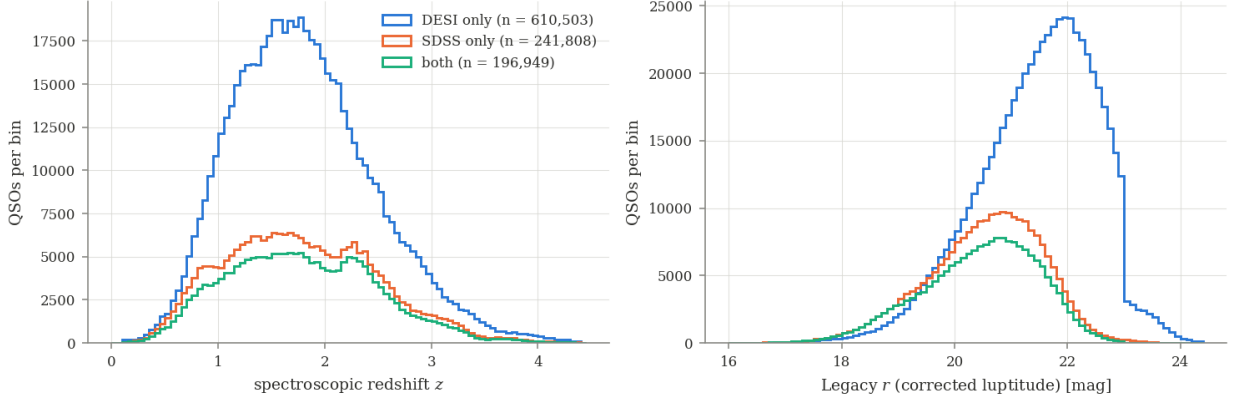


Figure 4: Fit/select quasars by parent survey. Left: spectroscopic redshift. Right: Legacy r (extinction-corrected luptitude, southern band where observed, otherwise northern). The DESI-only sample ends near DESI’s quasar target limit, $r \approx 23$.

5 Photometry and the observation model

5.1 Native luptitudes and measurement errors

Every band keeps its native calibration and zero point (AB for SDSS, Legacy, PS1, NSC and SkyMapper; Vega for AllWISE and VHS). Fluxes become luptitudes with a per-band softening s , and their variances propagate through the exact derivative of the luptitude function. A band that is not observed is marginalised exactly; it is never imputed.

5.2 Galactic extinction

Fluxes and variances are corrected before any other step:

$$f \rightarrow f 10^{0.4R_j E}, \quad \sigma_f^2 \rightarrow \sigma_f^2 10^{0.8R_j E}, \quad (4)$$

where E is the SFD98 reddening (Schlegel et al., 1998) at the source position. This preserves signal-to-noise and keeps negative fluxes negative. Table 1 lists the coefficients. Legacy DR9 values come from the survey documentation (Legacy Surveys, 2020), SDSS and PS1 values from Schlafly & Finkbeiner (2011), Table 6, and SkyMapper values from Wolf et al. (2018). NSC VR and VHS Y have no published values in these sources. We computed them with the Schlafly & Finkbeiner (2011) method: an F99 law with $R_V = 3.1$ and a 7000 K source, with one scale factor fitted to 16 published rows (residual rms 0.73%). Top-hat bandpasses were used, so these two values are approximate; we estimate $\pm 5\%$ for VR . The scorer computes E from the candidate’s position and applies the same correction, so users supply catalogue fluxes as observed. Fig. 5 shows that our correction reproduces the Legacy catalogue correction (`mw_transmission`) to 0.002 mmag.

5.3 One latent model for Legacy North and South

Legacy DR9 South (DECam) and North (BASS/MzLS) g, r, z use different filters. So that the northern fits can borrow from the much larger southern sample, we describe every source by latent, noise-free luptitudes $\mathbf{x}$ in 36 coordinates: five shared Legacy coordinates (`legacy:g,r,z,w1,w2`) and 31 coordinates for the other surveys. The 41 native bands follow from a linear operator,

$$\mathbf{y} = H\mathbf{x} + \mathbf{b} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim N(0, T). \quad (5)$$

Rows of H for the southern Legacy and all non-Legacy bands are unit vectors. Rows for the northern g, r, z carry the DESI North→South colour relation. For example, $y_{g,N} = 1.0627 x_g -$

Table 1: Extinction coefficients $R_j = A_j/E(B - V)_{\text{SFD}}$.

Bands	R_j	Source
Legacy DECam g, r, z	3.214, 2.165, 1.211	Legacy DR9 documentation
Legacy North g, r, z	3.278, 2.189, 1.196	DECam values through the DESI relation, Eq. (6)
W1, W2 (Legacy, AllWISE); W3, W4	0.184, 0.113; 0.0241, 0.00910	Legacy DR9 documentation
SDSS $ugriz$	4.239, 3.303, 2.285, 1.698, 1.263	Schlafly & Finkbeiner (2011), $R_V = 3.1$
PS1 $grizy$	3.172, 2.271, 1.682, 1.322, 1.087	Schlafly & Finkbeiner (2011)
NSC $ugrizY$	3.995, 3.214, 2.165, 1.592, 1.211, 1.064	DECam (Legacy DR9)
NSC VR	2.28	computed, approximate
SkyMapper $uvgriz$	4.294, 4.026, 2.986, 2.288, 1.588, 1.206	Wolf et al. (2018)
VHS $J, H, K_s; Y$	0.709, 0.449, 0.302; 1.00	UKIRT rows of Schlafly & Finkbeiner (2011); Y computed

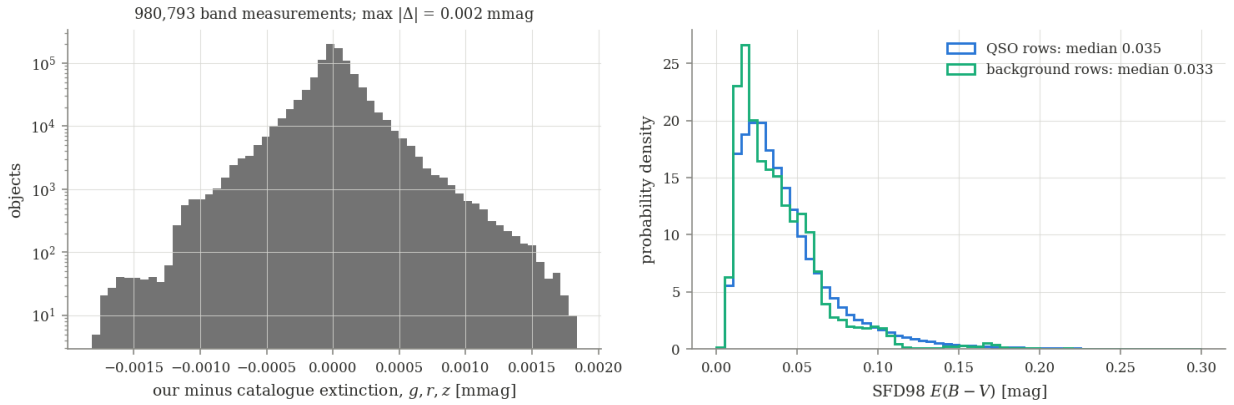


Figure 5: Left: our correction minus the Legacy catalogue correction, in four reserved Legacy cones (980,793 g, r, z measurements). Right: distribution of SFD98 reddening for the quasar and background rows (medians 0.035 and 0.033 mag).

$0.0643 x_r + 0.0016 x_z - 0.0060$. T holds the instrumental scatter measured on overlap pairs. The relation acts on uncorrected magnitudes, so extinction is removed in the DECam system:

$$R_{\text{native}} = H R_{\text{latent}}, \quad (6)$$

where each latent coordinate takes the coefficient of the system it represents. Fig. 6 shows that the relation removes the colour-dependent North–South offsets on the same objects. The Legacy $W1, W2$ coordinates are shared after both hemispheres use common units and softening; they remain distinct from AllWISE $W1, W2$.

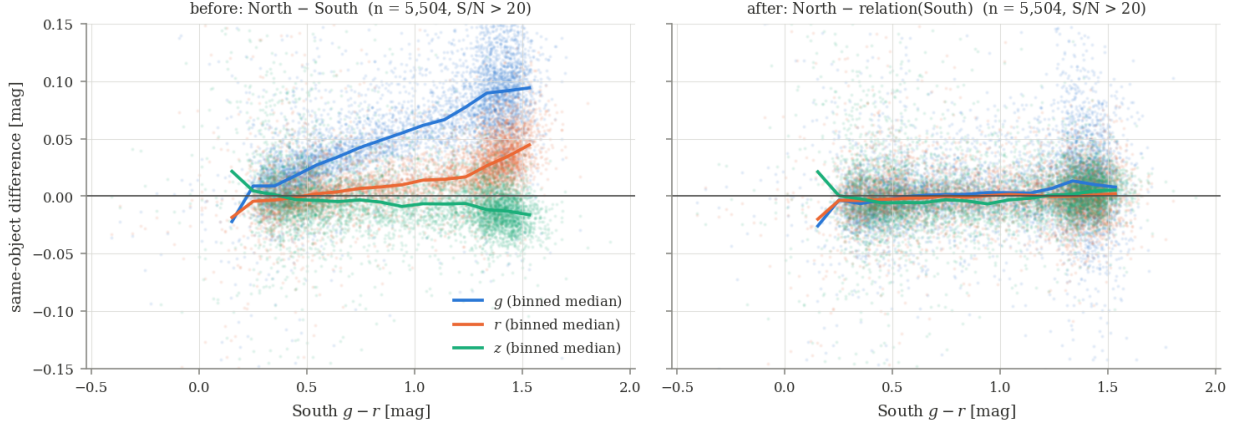


Figure 6: Same-object North minus South luminosities in overlap fields ($S/N > 20$ in all six bands), against South $g - r$. Left: raw difference. Right: after applying the DESI relation to the southern values. Lines: binned medians.

6 The quasar model

6.1 Redshift slices and extreme deconvolution

For each of the 43 slices, the latent quasar luminosities follow a Gaussian mixture with K components ($K = 20$ for 23 slices, 12 for 6, 8 for 13, and 4 for the sparsest one, 640 objects at $z = 4.35$):

$$p(\mathbf{x} \mid Q, z_k) = \sum_{i=1}^K \alpha_i N(\mathbf{x}; \boldsymbol{\mu}_i, V_i). \quad (7)$$

Extreme deconvolution (Bovy et al., 2011a) fits $\alpha_i, \boldsymbol{\mu}_i, V_i$ to noisy, incomplete data. Each object contributes $N(\mathbf{y}; H\boldsymbol{\mu}_i + \mathbf{b}, HV_iH^T + T + \mathbf{C})$ in its observed bands only, so measurement errors are removed from the fitted widths rather than absorbed into them. Section 8 describes the fitting. At scoring time the candidate’s own covariance is added back, and the density is conditioned on the reference band: $p(\mathbf{y}_{\text{other}} \mid m_{\text{ref}}) = p(\mathbf{y})/p(m_{\text{ref}})$. The scorer interpolates linearly between neighbouring slices in redshift.

6.2 Two quasar models: with and without magnitude dependence

We deliver two complete models. They are identical in everything (bands, observation model, extinction, background, catch-all, priors, faint limit and support procedure) except one assumption about quasars: whether a quasar’s colours at fixed redshift may depend on its apparent magnitude. Neither is a default.

Conditioning a joint mixture on m_{ref} lets a model’s colour distribution depend on apparent magnitude: a component’s colour mean and width shift with m_{ref} unless its covariance forbids it. Apparent magnitude enters the likelihood in both models through the abundance prior $\Sigma_Q(z, m)$; the question is only whether it also changes the predicted colours.

Magnitude-independent model. This is the XDQSO assumption (Bovy et al., 2011b). We rewrite the latent vector as one magnitude and 35 colours, $u_r = x_r$ and $u_j = x_j - x_r$ for $j \neq r$, with r the shared Legacy r coordinate. In every component the magnitude coordinate has the same fixed distribution, $N(20, 100 \text{ mag}^2)$, and zero covariance with the colours, so conditioning on the reference band leaves the colour likelihood independent of brightness (the residual dependence through the finite magnitude variance is below 10^{-3} nat for a $\pm 3 \text{ mag}$ shift). With this block structure the likelihood factorises, and the exact constrained M step is the ordinary update of the colour block (Appendix B); a unit test confirms that it equals a fit to the colours alone.

Magnitude-dependent model. The same mixture without the constraint: magnitude and colours are fitted jointly, so each slice’s colour distribution may shift and change width with brightness.

What each assumes, and its risk. The training quasars were selected by DESI and SDSS with colour cuts that change with magnitude, and quasar populations themselves change with luminosity. The magnitude-dependent model can describe any real dependence, but it can also learn the selection as if it were quasar physics and then penalise a genuine quasar whose colours fall outside the selection at its magnitude. The magnitude-independent model cannot learn that selection, but it must use one colour width for all magnitudes, so where the observed colour scatter changes with brightness it is too broad for bright quasars and too narrow for faint ones (Section 10.1). Our test quasars share the training selection, so the tests measure the cost of the magnitude-independent assumption on selected quasars, not its benefit for quasars outside the selection.

Measured differences (Section 10; held-out rows, faint limit applied; density of the magnitude-dependent model relative to the magnitude-independent one):

quantity	mag.-independent	mag.-dependent
held-out quasar density, per slice	—	+0.67 nat (0.38–0.88)
quasars inside the 50%/90% contours, $17 \leq r < 19$	0.62/0.94	0.51/0.90
quasars inside the 50%/90% contours, $21.5 \leq r < 22.5$	0.41/0.86	0.49/0.89
AUC, all magnitudes	0.976	0.980
AUC, $r < 18$	0.955	0.975
AUC, $18 \leq r < 19$	0.979	0.990
AUC, $r > 21$	0.969–0.973	0.970–0.978
AUC at $z_0 = 3.45$	0.948	0.956
quasars with $p_{\text{quasar}} > 0.5$	86%	89%
unflagged background with $p_{\text{quasar}} > 0.5$	0.28%	0.25%
quasars not ranked	3.2%	2.9%
grid points with $p_{\text{quasar}} > 0.5$, no support cut	12	4

On selected quasars the magnitude-dependent model is better on every measure, most clearly for bright quasars; the two converge at $r > 21$, where most quasars are. The magnitude-independent model also relies more on the support cut to avoid extrapolation.

Which to use. For candidates that resemble the training quasars, the magnitude-dependent model gives the better-measured performance. For candidates that may lie outside the DESI and SDSS selection, for example quasars with unusual colours for their brightness, the magnitude-independent model makes the weaker assumption. We recommend scoring every candidate with both: a candidate ranked highly by both is robust to this assumption, and a large disagreement between them marks a case that depends on it.

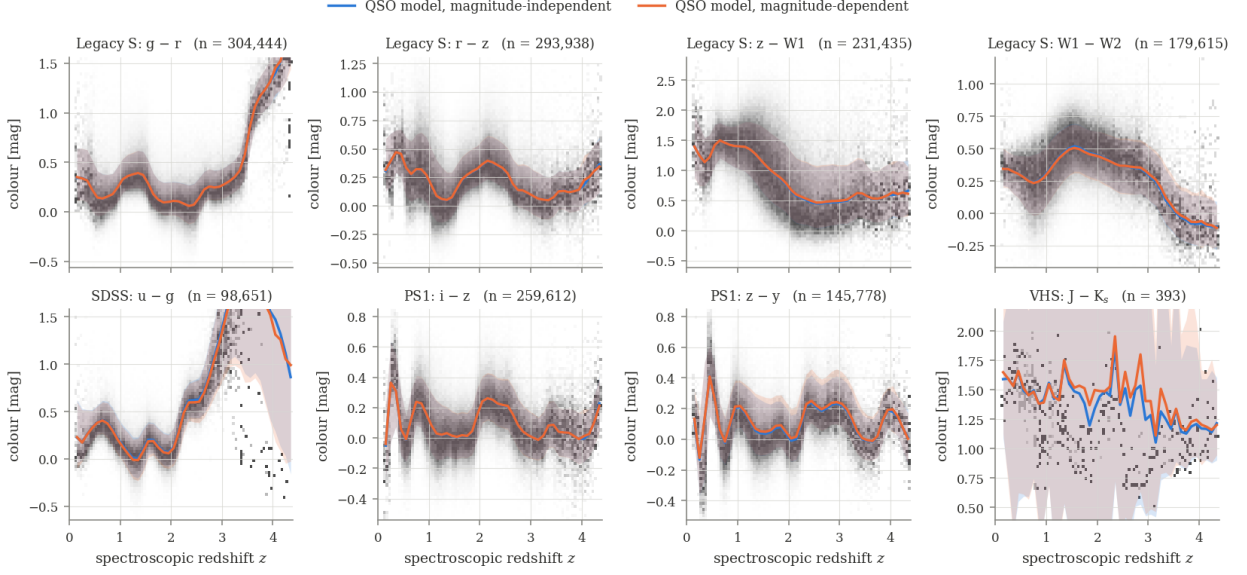


Figure 7: Quasar colour against redshift. Grey: test quasars with both bands at $S/N > 10$ (density normalised in each redshift column of width 0.05). Lines and bands: per-slice median and 16–84% range of the noise-free model colour, magnitude-independent (blue) and magnitude-dependent (orange, marginalised over magnitude). The data include measurement noise and a signal-to-noise selection that the model bands do not.

7 The background model

Design. Faint sources dominate the background counts (69% of fit/select rows have $r > 23.5$). A single mixture over all magnitudes therefore spends almost all its components on faint stars and describes bright ones poorly. Following XDQSO (Bovy et al., 2011b), which fits stars in narrow magnitude bins, we fit the background separately in bins of Legacy r . On held-out stars this describes the data better than one 20-component mixture over all magnitudes in every bin, by 4.9 nat per object at $r < 17$ and 0.2 at $r > 23$. Sharing one set of colour shapes across magnitude, with only their weights free, is worse at $r > 22$: the shape of the stellar locus itself changes with magnitude. CondXD (Kang et al., 2025) reaches comparable results with a neural network that makes all mixture parameters functions of magnitude.

Binned fit. The fit/select background rows above the faint limit are split into eight bins of Legacy r : 15.5–17, 17–18, ..., 22–23 and 23–24.5. Each bin is fitted independently by extreme deconvolution with $K = 20$ components over the 36 latent coordinates (written as r and 35 colours), on its own rows plus those within 0.25 mag of its edges (19,158 to 255,709 rows per bin), with the MAP update and predictive early stopping of Section 8 on 1,000 test-role objects per bin, starting from equal weights. The eight mixtures are joined into one 160-component mixture, each bin’s weights scaled by its share of training objects. Conditioning this mixture on m_{ref} weights each component by $N(m_{\text{ref}}; \mu_{r,i}, V_{rr,i})$, which selects the bin and blends neighbouring bins smoothly at the edges. On 17,937 held-out stars this beats picking the single bin that contains m_{ref} by 0.12 ± 0.015 nat per object overall (0.24 ± 0.07 within 0.1 mag of a bin edge, 0.08 ± 0.01 further than 0.3 mag), in both hemispheres and in every bin (from $+0.002 \pm 0.005$ at $22 \leq r < 23$ to $+0.23$). Because m_{ref} is Legacy r for every scored source, routing and conditioning use the same magnitude.

Training sample cleaned of quasars. Known quasars (DESI DR1, DR16Q) were removed from the background sample when it was built, but it still contains unidentified ones, increasingly at faint magnitudes: among background rows detected in $W1$ and $W2$, the fraction with the AGN colour $W1 - W2 > 0.8$ (Vega; Stern et al. 2012) is 3% at $19 \leq r < 20$ and 37% at $22 \leq r < 23$.

A binned model is flexible enough to learn them as background, which lowers the recall of real quasars (in a test, from 0.83 to 0.74 at $p_{\text{quasar}} > 0.5$). We therefore remove from training (and from the background test sets) members of Quiaia (Storey-Fisher et al., 2024) matched by Gaia DR3 `source_id`, and sources with $W1 - W2 > 0.8$ (Vega) at $S/N \geq 5$ in both bands, except very red optical sources ($r - z > 2$, kept as brown-dwarf candidates). This removes 0.1% of training rows at $r < 18$ and 3–5% at $r > 20$. A Gaia zero-proper-motion cut was tested and rejected: 33–56% of zero-motion sources have stellar $W1 - W2$ even at $\sigma_\mu < 0.2\text{--}0.5 \text{ mas yr}^{-1}$ (distant halo stars), and Gaia does not reach the faint bins where contamination matters. Faint unidentified quasars not detected by WISE remain; we accept this.

Sky dependence. Component shapes are the same everywhere; the weights vary with position as a softmax gate,

$$w_i(\ell, b) \propto \exp[a_i + \beta_i \cdot \mathbf{f}(\ell, b)], \quad \mathbf{f} = (\csc |b|, \cos \ell, \sin \ell, \csc |b| \cos \ell, \text{sign } b), \quad (8)$$

fitted by maximum likelihood with fixed component densities and an L2 penalty of 1 on β , on 300,000 cleaned training rows. The gate is fitted to the joint density, but it is judged by the quantity the scorer uses: on 60,000 held-out rows in the test cones, with the final background, it improves the conditional density $\ln p(\mathbf{y}_{\text{other}} | r)$ over one all-sky weight vector by 0.0260 ± 0.0015 nat per object, positively in every magnitude and $|b|$ bin, and beats a HEALPix hierarchy of cell weights (inside 4 pooled to inside 2 and all-sky, $n_0 = 1000$: 0.0219 ± 0.0011). The scorer evaluates the gate at inside-16 pixel centres (3.7° ; no measurable loss). The effect is small because stellar colours at $|b| \geq 25^\circ$ change little with position.

Surface density. $\Sigma_B(m)$ is measured per reference band from counts in regions of measured usable area, pooled with $n_0 = 100$, on the cleaned sample, so that counts and colour model describe the same population (removing the flagged sources lowers it by less than 0.5% at $r < 18.5$ and by 3–6% at $20 \leq r \leq 24.5$). A candidate-local refit of weights and density in a declared cone is available.

8 Fitting

8.1 The covariance update

The extreme-deconvolution E step computes each component’s responsibility q_{ni} for object n , and the posterior moments of $\mathbf{x}$. The M step updates the weights, the means, and the covariance from the scatter S_i about the new mean. We use the conjugate-prior MAP update of Bovy et al. (2011a) (their eqs. 19–20):

$$V_i = \frac{q_i S_i + \nu w I}{q_i + \nu}, \quad q_i = \sum_n q_{ni}, \quad \nu = 1, \quad w = 10^{-3} \text{ mag}^2, \quad (9)$$

which maximises the log posterior with prior $-\frac{\nu}{2} [\ln |V_i| + w \text{tr}(V_i^{-1})]$ per component, so EM cannot decrease the reported objective. (A fixed additive floor, $V_i = S_i + wI$, maximises no objective, and in practice lets the likelihood decline.)

8.2 Predictive early stopping

Every 20 updates we compute the held-out density on a fixed stopping panel: up to 512 test-role objects per hemisphere per fit, recorded and excluded from later validation. A fit stops when a block gains less than 0.02 nat per object, when the objective changes by less than 10^{-4} (relative), or after 300 updates. The best checkpoint, including the warm start, is kept. The 0.02 nat threshold is about twice the sky-bootstrap standard error of late block gains. This is predictive early stopping,

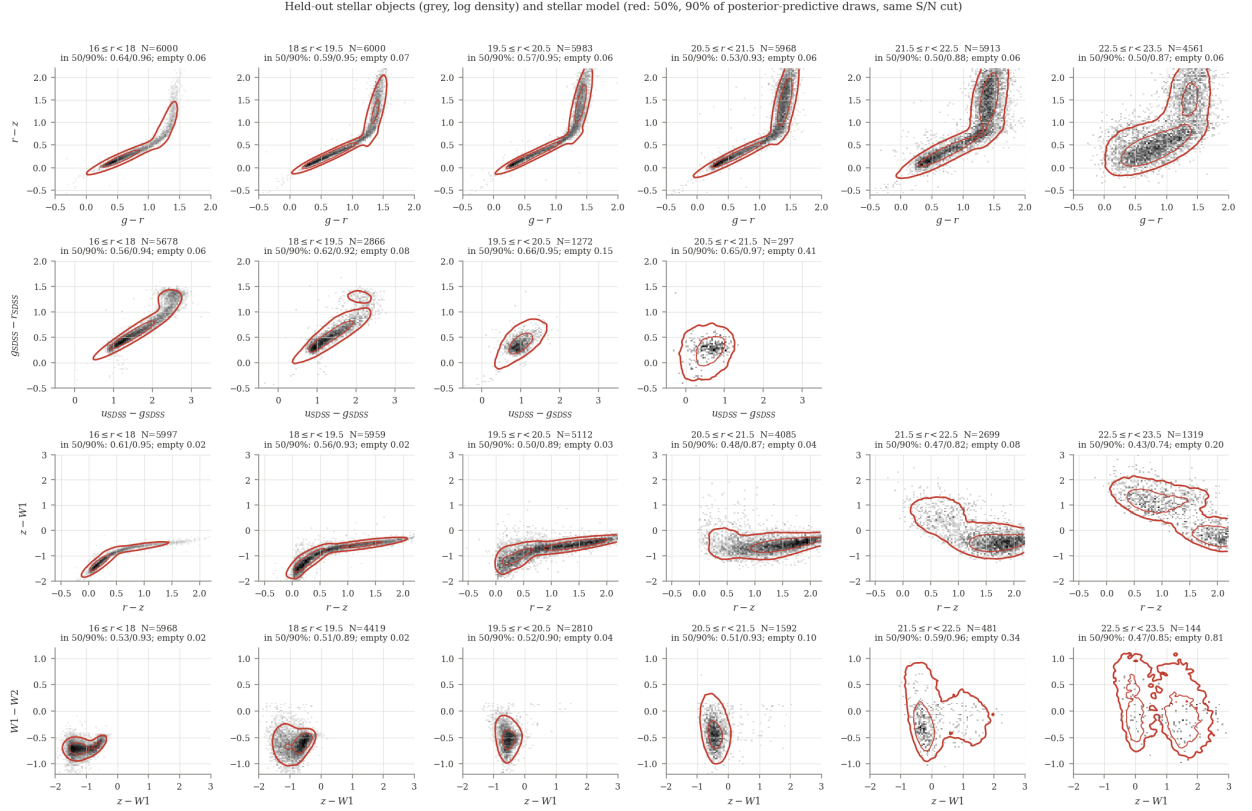


Figure 8: Background colour-colour density in six bins of Legacy r and four planes. Grey: test-cone background objects above the faint limit, likely quasars removed, all plane bands at $S/N \geq 5$. Red: 50% and 90% contours of posterior-predictive draws of the binned background for the same objects (200 per object, conditioned on each object's r and sky position, with the object's own flux errors added in flux space and the same $S/N \geq 5$ selection). Titles: data inside the 50%/90% regions (ideal 0.50/0.90) and the fraction of draws in empty cells. Panels with fewer than 100 objects are left empty.

not convergence. Sparse slices benefit most: where held-out density falls in the first block, as at $z = 4.05$, the fit keeps its warm start. All 43 quasar fits of each model stopped this way, with no objective decrease (Fig. 9a,b). The eight background bins use the same rule on 1,000 test-role objects per bin; they kept checkpoints at 40–92 updates (Fig. 9c).

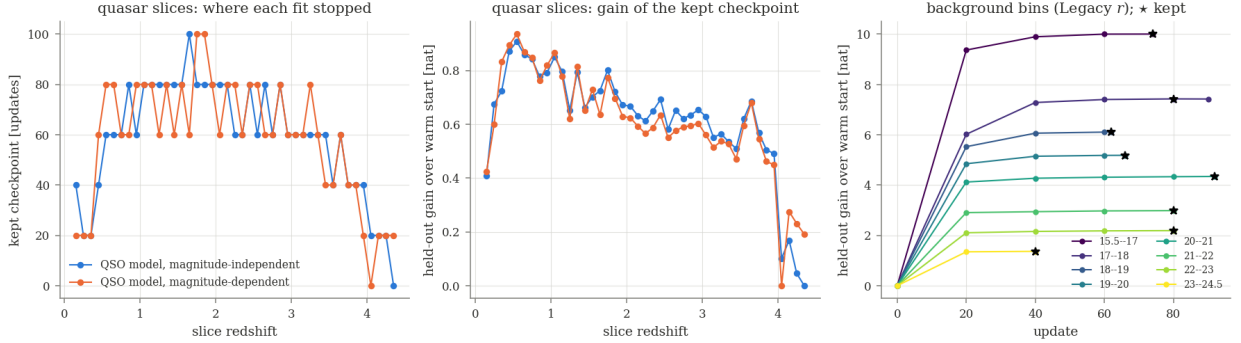


Figure 9: (a) Update at which each quasar-slice fit kept its checkpoint, against slice redshift. (b) Held-out gain of that checkpoint over the fit’s warm start. (c) Held-out density gain over the warm start against update for the eight background bins (Legacy r); stars mark the kept checkpoints.

9 Priors, catch-all and support

Abundance priors. $\Sigma_Q(z, m)$ and $\Sigma_B(m)$ are stored per reference band, per unit native luminosity. Background densities are counted from corrected photometry (Section 7). The quasar abundance comes from earlier full-sample counts on uncorrected photometry, moved onto corrected magnitudes by the mean extinction shift (0.080 mag in Legacy r , at mean $E = 0.037$); this is an approximation, not a recount.

Catch-all. U is a Student- t with $\nu = 4$ degrees of freedom, centred on the background mean, with the binned background mixture’s second-moment matrix as its scale. Measurement noise is convolved exactly through the t ’s Gaussian scale-mixture representation. Its share η is a Beta-MAP fit per reference band and three magnitude bins, on 148,853 calibration background objects above the faint limit with likely quasars removed. The global share is 0.14%.

Support cut. For a candidate, z_0 and the observed bands, the scorer draws 255 simulated quasars from the conditional model at z_0 , adds the candidate’s errors, and ranks the candidate’s conditional density among them. Candidates below a threshold are not ranked; their likelihoods are kept as diagnostics. The percentile depends on the reference band, so the threshold was recalibrated under the faint limit on 2,000 calibration quasars (all bands and Legacy g, r, z only; 3,890 evaluations). At the 99.5% target it is $2/256 = 0.0078$, retaining 99.64% and 99.59% of calibration evaluations. Fig. 10 shows the trade-off on held-out rows: the cut at $2/256$ removes all 12 (magnitude-independent) and 4 (magnitude-dependent) high-quasar points that the artificial grids place outside the models’ dense core, at a cost of 0.5% of test quasars, and raises the AUC of p_{quasar} ; stricter cuts lose quasars faster than they gain.

10 Validation

Tests use test-role objects above the faint limit and outside every recorded development exclusion.

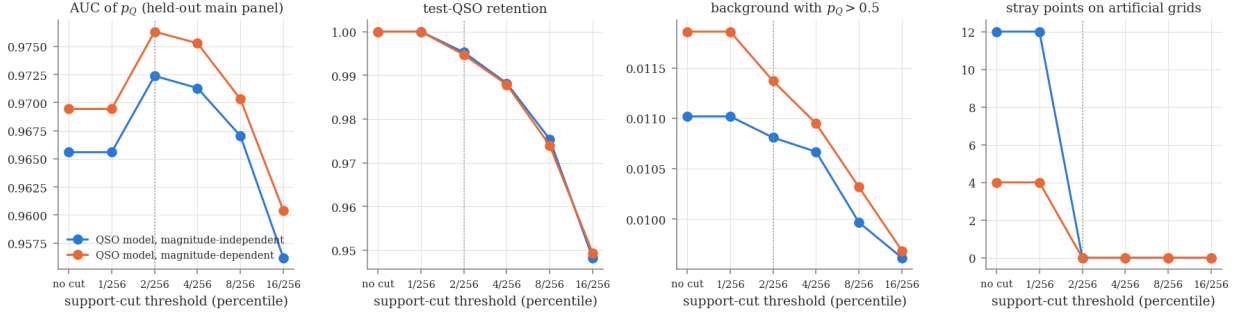


Figure 10: Support-cut threshold (percentile; dotted: the calibrated 2/256) on the held-out main panels of Section 10.3 and the artificial grids of Section 10.4: AUC of p_{quasar} (the scorer does not compute $\log \mathcal{R}$ for rejected objects, so p_{quasar} ranks all thresholds alike; unranked objects at the bottom), fraction of test quasars kept, fraction of background objects with $p_{\text{quasar}} > 0.5$, and grid points with $p_{\text{quasar}} > 0.5$ in the low-density mask.

10.1 Colour–colour distributions

Figures 11–13 compare model densities with the data in two colour planes: Legacy South ($g-r, r-z$) and ($r-z, z-W1$). Model contours are conditioned on each bin’s median r , weighted by the data’s redshift mix, and convolved with the bin’s median measurement covariance, so they can be compared directly with the data. Each panel title gives the fraction of the data inside the model’s 50% and 90% regions; a well-calibrated model gives 0.50 and 0.90. The magnitude-dependent quasar model gives 0.48–0.51 and 0.89–0.90 in every magnitude bin and both planes. The magnitude-independent model must use one width for all magnitudes. It is too broad for bright quasars (0.62/0.94 at $17 \leq r < 19$) and too narrow for faint ones (0.41/0.86 at $21.5 \leq r < 22.5$). The apparent colour scatter of quasars, after deconvolution, grows towards faint magnitudes in this sample. Underestimated photometric errors, a change in the population, and selection can all produce this. The binned background follows the stellar locus at every magnitude and sky position: in ($g-r, r-z$) it gives 0.43–0.51 and 0.89–0.92 in the magnitude bins and 0.42–0.46 and 0.89–0.90 in the sky bins. In ($r-z, z-W1$) the 90% coverage is 0.82–0.88; part of this is the $W1$ detection selection of the plotted data, which Fig. 8 reproduces in the model and which brings the coverage to 0.87–0.95 for $r < 21.5$.

10.2 Held-out density

Per redshift slice, the magnitude-dependent quasar model’s held-out density is higher than the magnitude-independent model’s by a median 0.666 nat per object (range 0.381–0.876): the cost of the magnitude constraint in density (Section 6.2). The background’s fit is shown directly in Fig. 8.

10.3 Performance across sky, magnitude, redshift and band coverage

These tests use 20,000 quasars and 20,000 background objects per hemisphere, drawn before the faint limit from test-role rows outside every development exclusion and outside the rows used to choose support thresholds; 38,881 quasars (97.2%) and 14,247 background objects (35.6%) pass the faint limit. Background objects are scored at redshifts drawn from the test quasars’ redshifts. The four metrics are those of Section 2: AUC, held-out density, support retention, and high-QSO incidence, the fraction of background objects ranked with $p_{\text{quasar}} > 0.5$. The background test objects include unidentified quasars, so incidence is not a contamination rate.

Overall. The AUC is 0.976 (magnitude-independent) and 0.980 (magnitude-dependent); South 0.974 and 0.978, North 0.978 and 0.982. The support cut keeps 99.53% and 99.47% of test quasars, in agreement with the 99.5% calibration target on rows that did not choose it. In total 3.2% and 2.9% of test quasars are not ranked: besides the support cut (0.25% and 0.30%), 2.9% and 2.5%

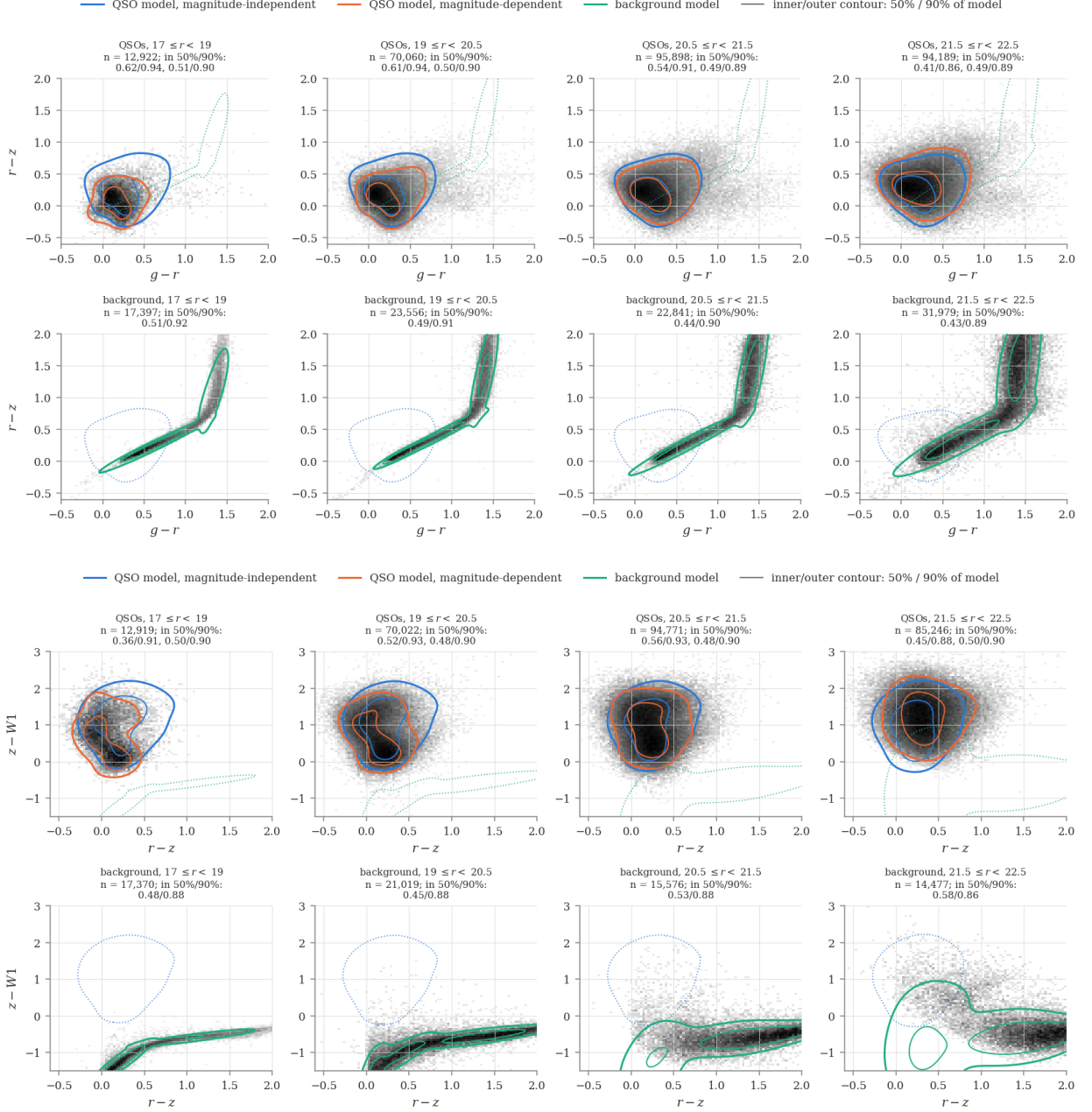


Figure 11: Colour-colour planes in bins of Legacy South r , objects above the faint limit. Top rows of each block: test quasars (grey, log density) with both quasar models (blue, orange; thick and thin contours enclose 90% and 50%) and the background model's 90% contour (dotted aqua). Bottom rows: test background objects (likely quasars removed) with the binned background and the magnitude-independent quasar model's 90% contour (dotted blue).

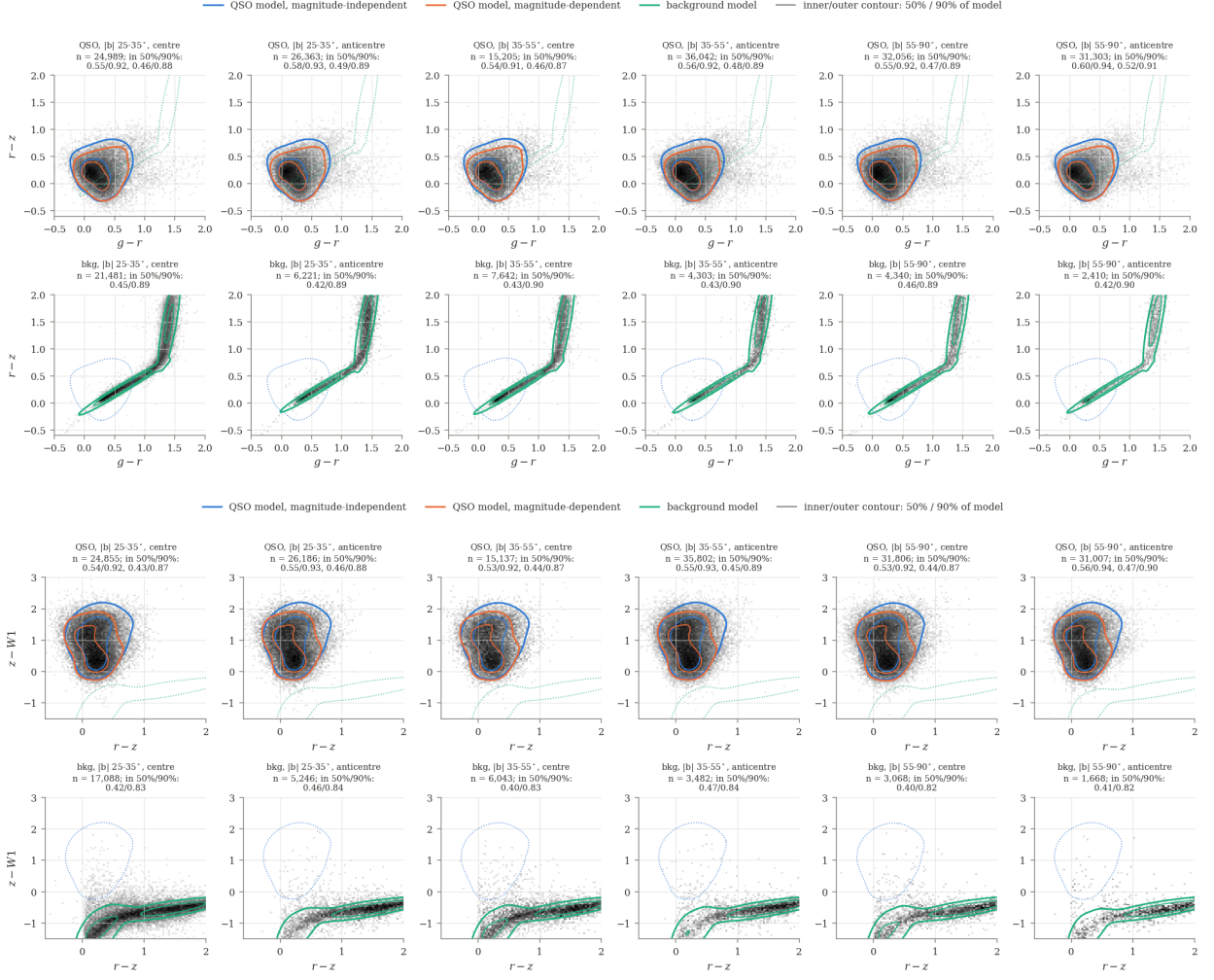


Figure 12: As Fig. 11, in bins of Galactic latitude and of longitude towards ($|l| < 90^\circ$) or away from the Galactic centre, at $19 \leq r < 21.5$. Background contours use the spatial component weights averaged over each bin's objects.

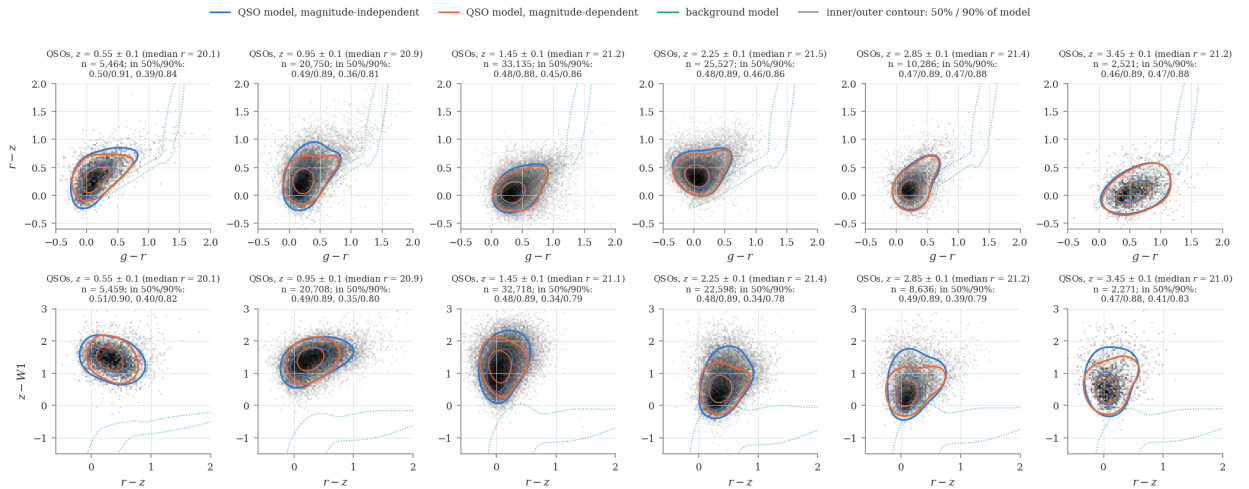


Figure 13: Quasar colour-colour planes in six redshift slices, each conditioned on the slice's median r , with the background model's 90% contour for reference.

have colours outside both the quasar and the background model. Of the test quasars, 86% and 89% receive $p_{\text{quasar}} > 0.5$. High-QSO incidence is 1.08% and 1.14% (South 0.50% and 0.56%, North 1.76% and 1.81%). Among background objects that are not flagged as likely quasars it is 0.28% and 0.25%, and the AUC against them is 0.976 and 0.980.

Sky position and magnitude (Fig. 14). Per-cell AUC is uniform across the footprint to within a few thousandths. Against $|b|$, AUC is 0.973–0.981 (magnitude-independent) and 0.975–0.985 (magnitude-dependent), highest at the lowest latitudes. Against reference magnitude the two quasar models differ most for bright quasars: AUC 0.955 against 0.975 at $r < 18$ (274 quasars), and 0.979 against 0.990 at $18 \leq r < 19$, where the magnitude-independent model’s single colour width is too broad (Section 10.1). They converge at $r > 21$ (0.969–0.973 against 0.970–0.978).

Redshift (Fig. 15). AUC is 0.967–0.980 (magnitude-independent) and 0.976–0.983 (magnitude-dependent) for $0.5 \leq z_0 \leq 3$, and falls to 0.948 and 0.956 at $z_0 = 3.45$, where quasar colours cross the stellar locus. Held-out density falls with redshift as quasars become fainter, and the magnitude-dependent model stays higher at every redshift. Support retention is above 99% everywhere.

Band coverage (Fig. 16). Grouping objects by the surveys that observed them mixes coverage with brightness and redshift, and several groups have fewer than 150 background objects. The artificial masks score the same objects with fewer bands: Legacy g, r, z only gives AUC 0.979 and 0.980, all Legacy bands 0.993 and 0.994, all optical bands 0.990 for both. SDSS-only and PS1-only photometry has no Legacy r and is not scored. The weakest coverage group, SDSS+Legacy South+AllWISE+PS1+NSC (AUC 0.909 and 0.917, 347 quasars), lies in the NSC region of the southern sky.

10.4 Artificial colour grids

A colour grid places synthetic sources on a 15×15 lattice in $(g-r, r-z)$ from -6 to 8 mag at fixed r (18.5 and 21; South and North; $(\ell, b) = (180^\circ, 45^\circ)$; 0.03 mag errors). These are stress tests of extrapolation into empty colour space, not contamination rates. Outside the models’ dense core, where both the quasar and the background intensity are below 1% of their grid maximum, no grid point receives $p_{\text{quasar}} > 0.5$ with the support cut (Fig. 17); without it, 12 (magnitude-independent) and 4 (magnitude-dependent) would. Most of the grid is not ranked: points outside both models, or rejected by the support cut, are set aside.

10.5 Worked examples

Figure 19 shows six test objects scored by both models. For each band the panels compare the observed corrected magnitude, minus Legacy r , with the model’s conditional prediction at the object’s redshift. A bright $z = 1.51$ quasar, a $z = 3.00$ quasar and a $z = 0.58$ quasar match their predictions and receive $p_{\text{quasar}} = 1.00$ ($\log \mathcal{R} = 1.4$ – 1.6 , 0.5 – 0.6 and 1.5 – 1.6). One test quasar at $z = 2.26$ has a Legacy r 2–3 mag fainter than all its other optical bands, a faulty or variable reference measurement; relative to it the colours are impossible for a quasar, and both models decline to rank it (support percentile 0.00). A background star has $\log \mathcal{R} = -41.6$ (magnitude-independent); the magnitude-dependent model rejects it at the support cut. The last panel is a background object that both models call a quasar ($p_{\text{quasar}} = 0.99$) but not one at z_0 ($\log \mathcal{R} \approx -6$): its colours lie on the quasar locus at another redshift, and it is probably an unidentified quasar. Legacy and AllWISE $W1, W2$ appear as separate points because Legacy fluxes are AB and AllWISE Vega; the zero points differ by about 2.7 and 3.3 mag.

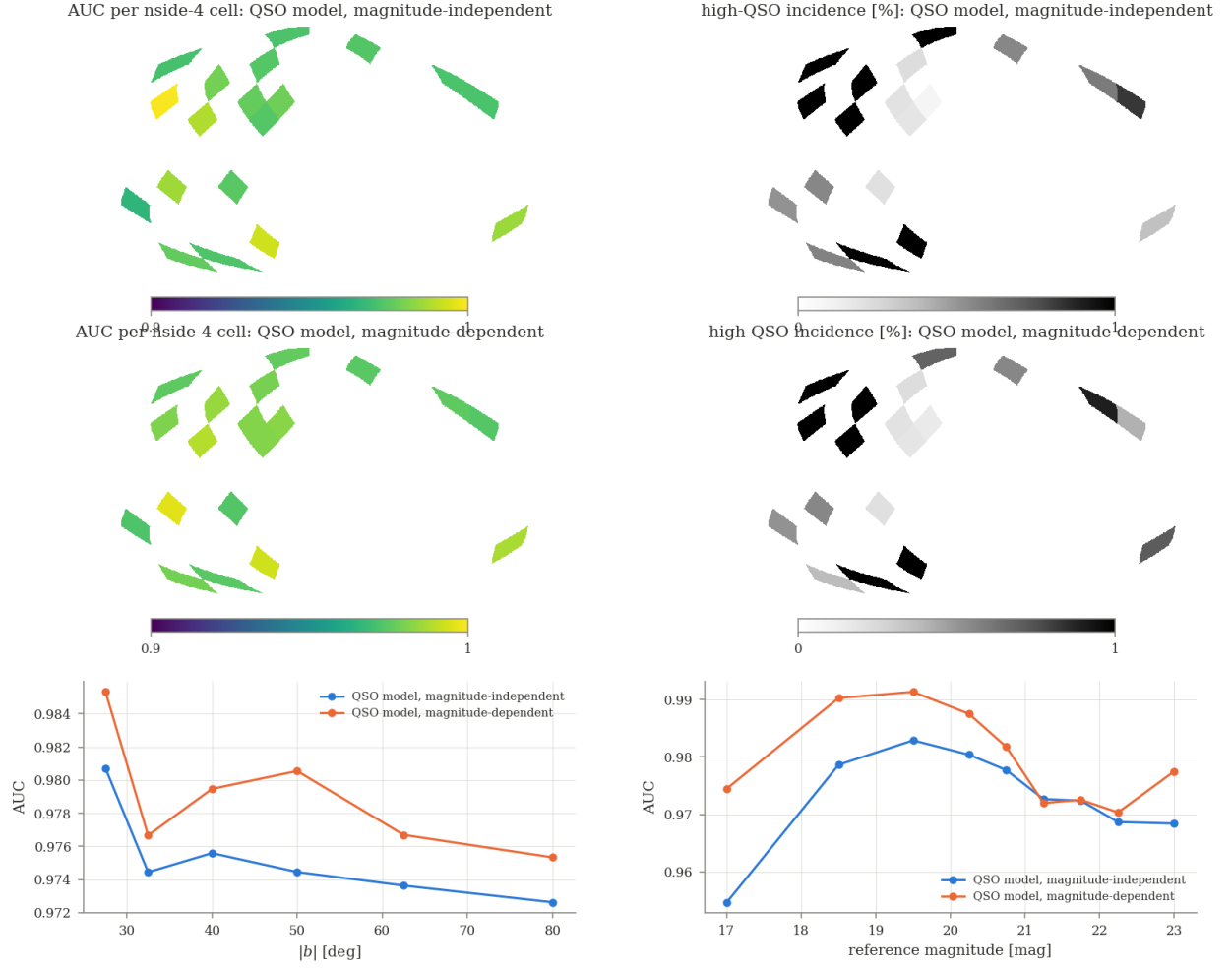


Figure 14: Performance against sky position and magnitude. Top: AUC and high-QSO incidence per HEALPix nside-4 cell (cells with at least 30 objects of each class). Bottom: AUC against $|b|$ and against reference-band magnitude.

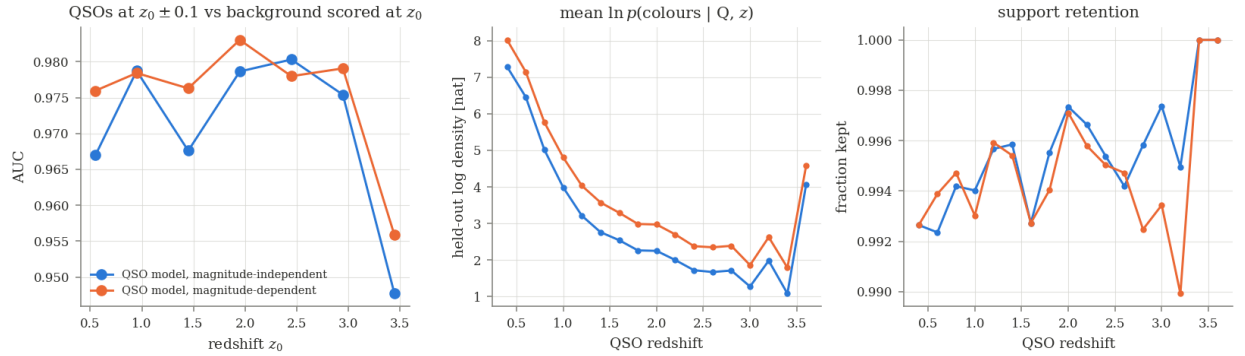


Figure 15: Performance against redshift. Left: AUC of quasars at $z_0 \pm 0.1$ against background objects scored at z_0 . Middle: mean held-out log density of quasars, in bins of their redshift. Right: support retention.

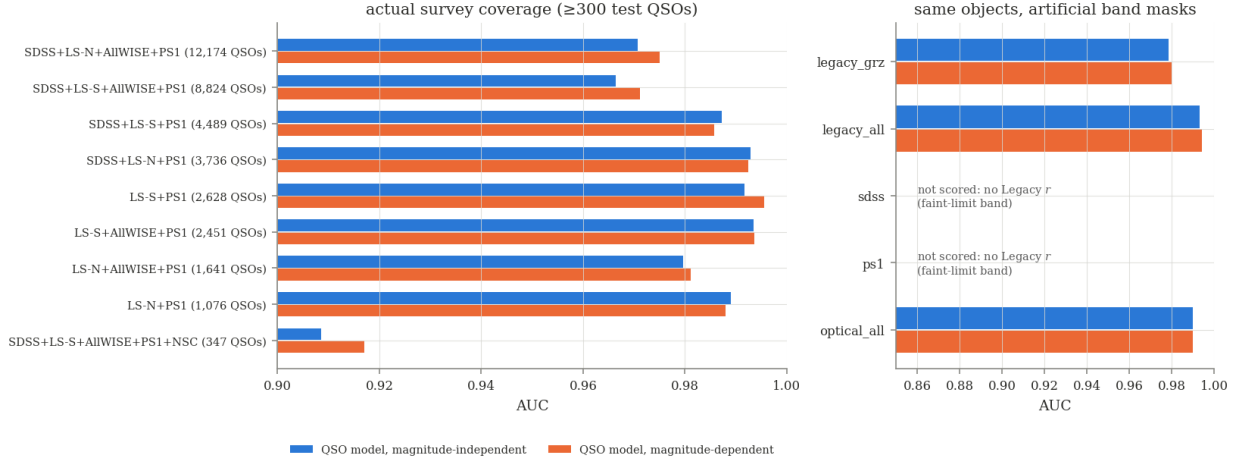


Figure 16: AUC by band coverage. Left: objects grouped by the surveys that observed them (groups with at least 300 test quasars). Right: the same objects scored with artificial band masks; SDSS-only and PS1-only objects have no Legacy r and are not scored.

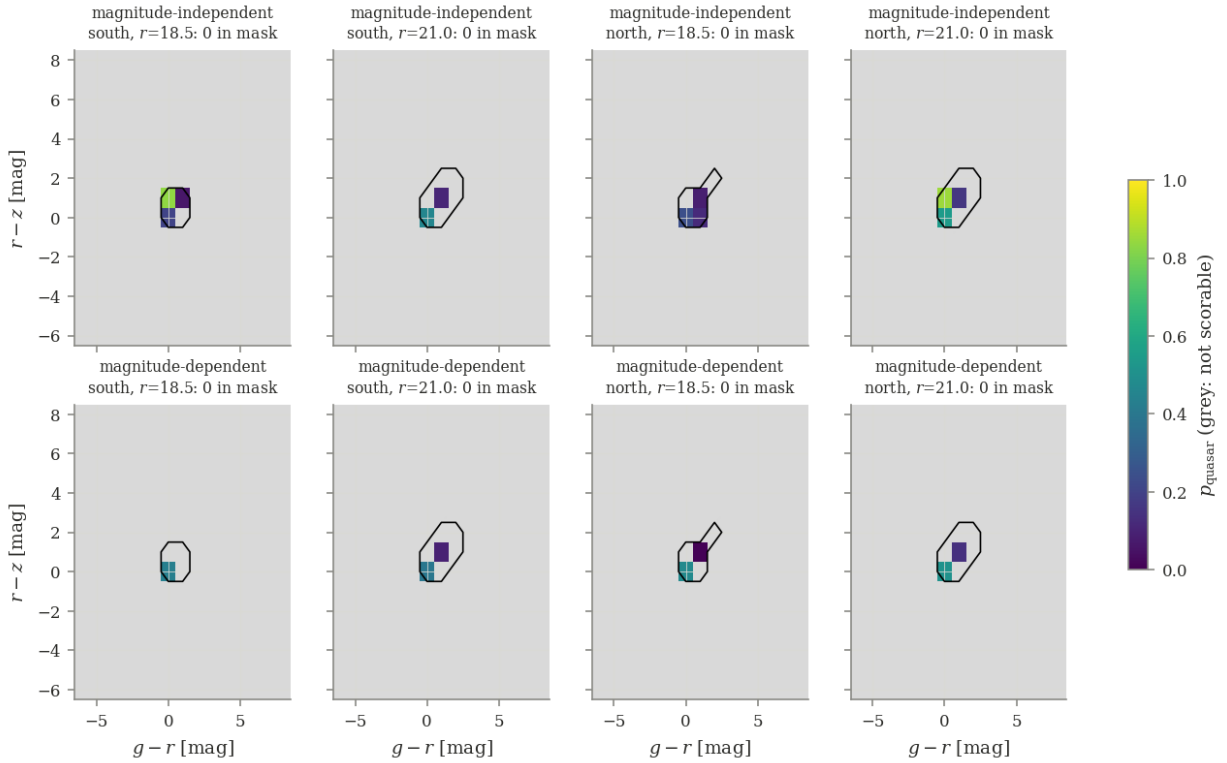


Figure 17: Total quasar probability on artificial Legacy colour grids, South and North, $r = 18.5$ and 21. Grey cells are not ranked. Black outline: the models' dense core; outside it both the quasar and the background intensity are below 1% of their grid maximum.

10.6 Probability calibration

We test whether p_{quasar} is a calibrated probability on a natural population: all 1,640 known spectroscopic quasars inside the 49 test cones that pass the faint limit, and 40,000 of the cones' background objects, weighted by 2.75 to restore their natural share (1.47% quasars). The background sample still holds unidentified quasars, so the observed quasar fraction is bracketed by two labellings: known quasars only (a lower bound), and known quasars plus flagged likely quasars (an upper bound).

We separate two questions. The first is abstention. The scorer declines to rank 3.5% and 3.8% of the known quasars (magnitude-independent and magnitude-dependent model), 53% and 63% of the background objects, and 36% and 41% of the flagged likely quasars. Background objects are set aside almost entirely by the support cut (96%); known quasars mostly because their colours lie outside both models (85%). The second is the reliability of the returned probability, among ranked objects only (Fig. 18): in the mid range, $0.1 \lesssim p \lesssim 0.95$, the observed quasar fraction brackets p_{quasar} in most bins (for example 0.30–0.66 at $p \approx 0.58$, and 0.48–0.87 at $p \approx 0.85$, magnitude-independent). Ranked objects at $p < 0.01$ contain no known quasars and 2.5% flagged ones. At the top the probability is too high: $p > 0.99$ gives 0.69–0.98 observed. A two-parameter map of the log-odds, fitted on half the cones, is driven by the large $p \approx 0$ population and pushes mid-range values below both bounds ($p = 0.5 \rightarrow 0.29$), so it is not applied. p_{quasar} is therefore an approximately calibrated probability for ranked objects whose largest values are mildly overconfident; an object the scorer declines to rank should be treated as not a quasar candidate, which loses 3.5–3.8% of real quasars in this sample.

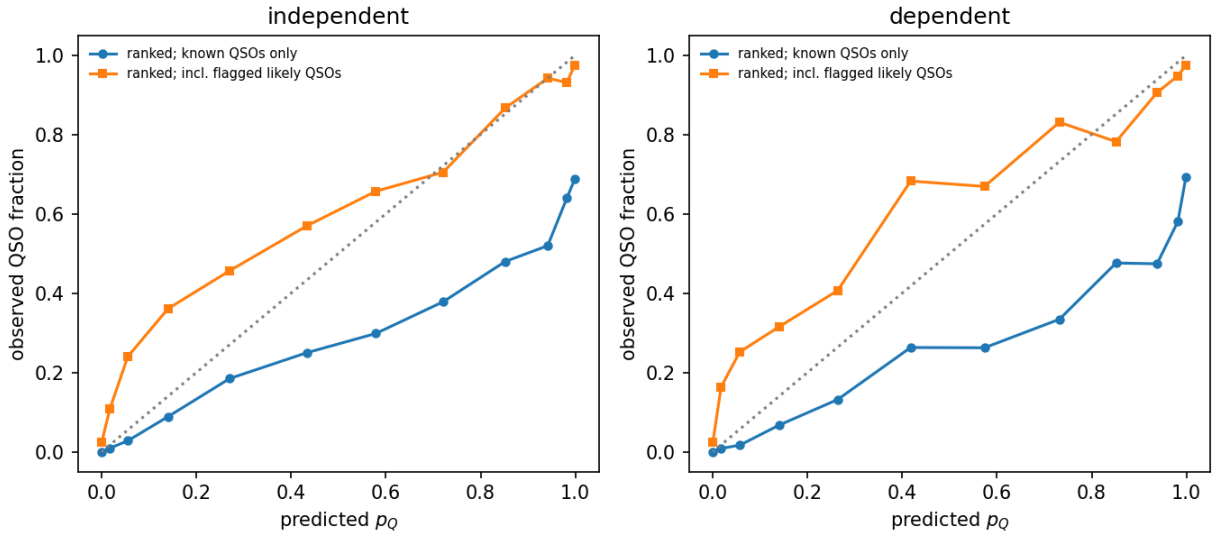


Figure 18: Reliability of p_{quasar} among ranked objects in the 49 test cones (natural population, weighted), for both models: observed quasar fraction against mean predicted p_{quasar} in bins, counting known quasars only (blue, lower bound) or also flagged likely quasars (orange, upper bound). Dotted: perfect calibration. Abstention rates are given in the text.

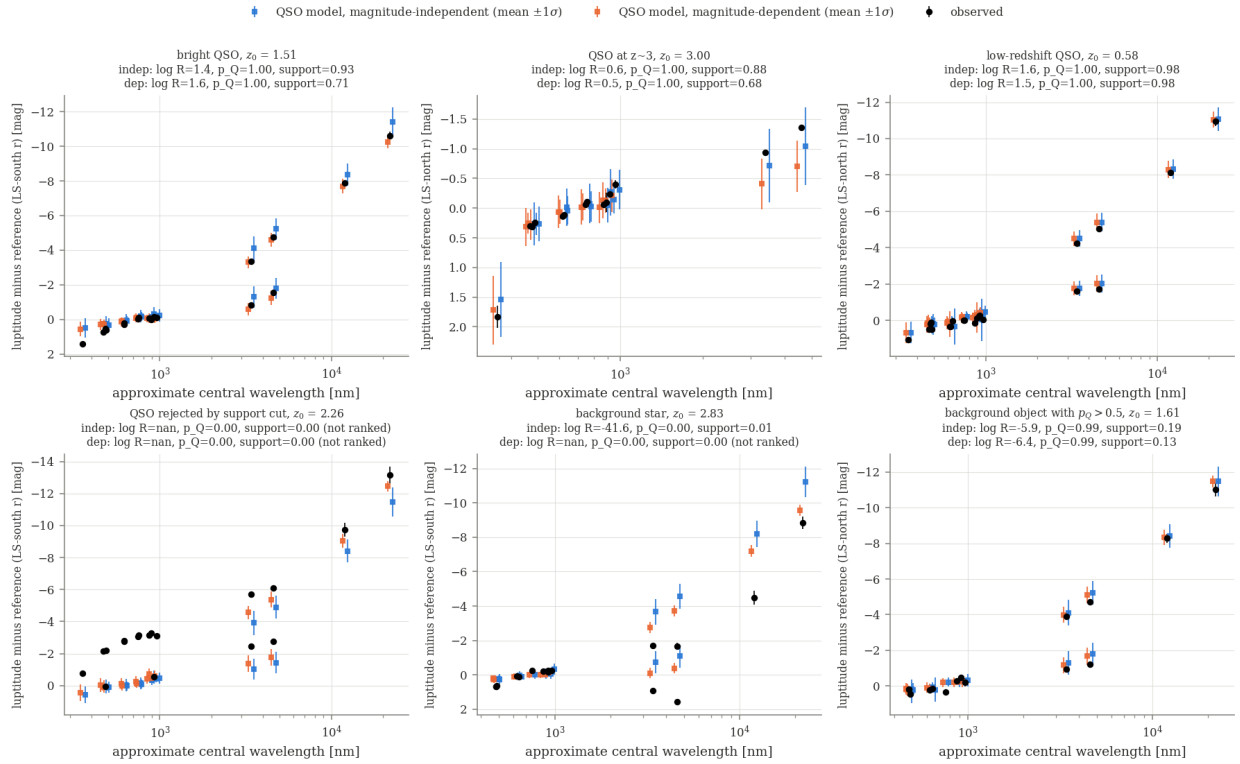


Figure 19: Worked examples: observed luminosity minus reference-band luminosity (black) and each model’s conditional prediction at the object’s redshift (mean $\pm 1\sigma$; blue magnitude-independent, or orange magnitude-dependent). Horizontal positions are approximate central wavelengths, for display only. Titles give $\log \mathcal{R}$, p_{quasar} and the support percentile from both models. Object identifiers are in `docs/method_unified/examples.json`.

11 Limitations and open items

- **Probability calibration** is approximate: among ranked objects p_{quasar} is bracketed by the label bounds in the mid range and overconfident above 0.97, and 3.5–3.8% of real quasars are not ranked (Section 10.6); the reliability of p_{same} for a narrow redshift window is not tested.
- **Background purity.** Faint unidentified quasars not detected by WISE remain in the background training sample and counts. The test background sets carry the same contamination, which lowers the measured AUC slightly.
- **Faint limit.** Sources below Legacy r $S/N = 10$, and sources without Legacy r , are not scored.
- **Unranked quasars.** About 3% of test quasars above the faint limit are not ranked because their colours lie outside both models; the cause is not yet investigated.
- **Selection.** Test quasars share the training selection. Our tests cannot measure the benefit of the magnitude-independent model for quasars outside the DESI and SDSS selection, so the choice between the two models (Section 6.2) rests partly on that untested risk.
- **Quasar abundance** comes from earlier counts shifted for extinction, not a recount.
- **Sparse surveys.** NSC VR and VISTA Y extinction coefficients are approximate, and SkyMapper u, v and VHS have little training support.
- **Background in the faintest bins** ($r > 21.5$) is broader than the data in $W1 - W2$ and slightly narrower in $r - z$ against $z - W1$ (Fig. 8); these panels hold few WISE-detected stars.
- **Out of scope.** Blends below the separation floor, and a physical-pair clustering prior, are deferred.

A The window factorisation

Write $\Lambda_Q = \int \lambda_Q(z) dz$ for the total quasar intensity, and let the declared window W around z_0 have effective width $\Delta z_{\text{eff}} = \int_W k(z) dz$ for its kernel k . The posterior that the companion is a quasar in the window is

$$p_{\text{same}} = \frac{\int_W k(z) \lambda_Q(z) dz}{\Lambda_Q + \lambda_B + \lambda_U} = \Delta z_{\text{eff}} \frac{\langle \lambda_Q \rangle_W}{\Lambda_Q + \lambda_B + \lambda_U} \equiv \Delta z_{\text{eff}} \mathcal{R}, \quad (10)$$

with $\langle \lambda_Q \rangle_W$ the kernel-weighted mean over the window. The denominator contains no window, because Λ_Q includes both same-redshift and other-redshift quasars. The scorer integrates the window on dense nodes at the support boundaries rather than on the coarse redshift grid. For a window much narrower than the photometric redshift width, $\langle \lambda_Q \rangle_W = \lambda_Q(z_0)$, which is Eq. (3).

B The constrained M step

For a component with mean $(\mu_m, \boldsymbol{\mu}_c)$ and covariance $\text{diag}(\sigma_m^2, V_c)$, where μ_m and σ_m^2 are fixed, the expected complete-data log likelihood separates into a magnitude term, constant in the free parameters, and a colour term. Maximising the colour term with the MAP prior on V_c gives the usual updates applied to the colour block alone. The prior on the full covariance differs from the prior on V_c by a constant. The constrained update is therefore exact.

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